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Java SE 27 & JDK 27

Interface ForkJoinPool.ManagedBlocker

Enclosing class:

`ForkJoinPool`

`public static interface` **`ForkJoinPool.ManagedBlocker`**

Interface for extending managed parallelism for tasks running in `ForkJoinPool`s.

A `ManagedBlocker` provides two methods. Method `isReleasable()` must return `true` if blocking is not necessary. Method `block()` blocks the current thread if necessary (perhaps internally invoking `isReleasable` before actually blocking). These actions are performed by any thread invoking `ForkJoinPool.managedBlock(ManagedBlocker)`. The unusual methods in this API accommodate synchronizers that may, but don't usually, block for long periods. Similarly, they allow more efficient internal handling of cases in which additional workers may be, but usually are not, needed to ensure sufficient parallelism. Toward this end, implementations of method `isReleasable` must be amenable to repeated invocation. Neither method is invoked after a prior invocation of `isReleasable` or `block` returns `true`.

For example, here is a ManagedBlocker based on a ReentrantLock:

```
class ManagedLocker implements ManagedBlocker {
  final ReentrantLock lock;
  boolean hasLock = false;
  ManagedLocker(ReentrantLock lock) { this.lock = lock; }
  public boolean block() {
    if (!hasLock)
      lock.lock();
    return true;
  }
  public boolean isReleasable() {
    return hasLock || (hasLock = lock.tryLock());
  }
}
```

Here is a class that possibly blocks waiting for an item on a given queue:

```
class QueueTaker<E> implements ManagedBlocker {
  final BlockingQueue<E> queue;
  volatile E item = null;
  QueueTaker(BlockingQueue<E> q) { this.queue = q; }
```

```
  public boolean block() throws InterruptedException {
    if (item == null)
      item = queue.take();
    return true;
  }
  public boolean isReleasable() {
    return item != null || (item = queue.poll()) != null;
  }
  public E getItem() { // call after pool.managedBlock completes
    return item;
  }
}
```

Since:

1.7

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
boolean	`block()`	Possibly blocks the current thread, for example waiting for a lock or condition.
boolean	`isReleasable()`	Returns `true` if blocking is unnecessary.

Method Details

block

```
boolean block()
       throws InterruptedException
```

Possibly blocks the current thread, for example waiting for a lock or condition.

Returns:

`true` if no additional blocking is necessary (i.e., if isReleasable would return true)

Throws:

`InterruptedException` - if interrupted while waiting (the method is not required to do so, but is allowed to)

isReleasable

```
boolean isReleasable()
```

Returns `true` if blocking is unnecessary.

Returns:

`true` if blocking is unnecessary

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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